Corey Oses

Materials Science and Engineering, Johns Hopkins University
Physics and Astronomy (joint), Johns Hopkins University

Work Experience · Education · Honors and Awards · Journal Publications · Book Publications · Talks/Presentations · Software and Datasets · Teaching Experience · Conference and Workshop Organization · Professional Service · Press and News Releases

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Work Experience

Assistant Professor	2022–present	Johns Hopkins University • Joint appointment, Department of Physics and Astronomy (2025–present)
Postdoctoral Fellow	2018–2022	Duke University

Education

Ph.D.	2013–2018	Duke University **Department**: Mechanical Engineering and Materials Science **Thesis**: *Machine learning, phase stability, and disorder with the Automatic Flow Framework for Materials Discovery* **DukeSpace**: hdl.handle.net/10161/18254
B.Sc.	2009–2013	Cornell University **Department**: Applied and Engineering Physics **Thesis**: *Plume Propagation Simulation for Pulsed Laser Deposition*

Honors and Awards

Award	2026	Finalist, Advanced Nuclear Energy Technology Challenge, Defense TechConnect
Award	2026	JHU Catalyst Early Career Award, Johns Hopkins University
Award	2026	MGB-SIAM Early Career Fellowship, Society for Industrial and Applied Mathematics
Award	2026	Early Career Award: Inspiring Generations of New Innovators to Impact Technologies in Energy 2025 (IGNIITE 2025), Advanced Research Projects Agency-Energy (ARPA-E)
Publication Award	2026	Most Popular Articles Collection, Publication in Mater. Horiz., Royal Society of Chemistry: Materials Horizons
Award	2025	Nexus Award, Johns Hopkins University
Award	2024	Early Career Investigator Award in Materials Modeling, International Society of Materials Modeling
Publication Award	2024	Editors' Highlight, Publication in Nat. Commun., Springer Nature
Award	2023	Reviewer of the Year 2022, npj Computational Materials
Publication Award	2022	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier

Publication Award	Nov 16, 2021	"Hot paper", Publication in Nat. Rev. Mater., Web of Science (Clarivate Analytics) • Published in the past two years and received enough citations in July/August 2021 to place it in the top 0.1% of papers in the academic field of Materials Science
Publication Award	2021	Editors' Highlight, Publication in Nat. Commun., Springer Nature
Publication Award	2018	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier
Publication Award	2017	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier
Award	Aug 14, 2015	Best Teaching Assistant Award (ME 221), Duke University Department of Mechanical Engineering and Materials Science
Publication Award	2015	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier
Publication Award	2015	Top 10 most highly downloaded papers for the month of January 2015, Publication in Chem. Mater., American Chemical Society
Publication Award	2015	Editors' Choice, Publication in Chem. Mater., American Chemical Society
Fellowship	2013–2016	Graduate Research Fellowship, National Science Foundation

Journal Publications † contributed equally * corresponding

2026

50. S.-Y. Tseng, G. Han, T. Li, X. Xu, J. Hu, G. Qiu & C. Oses*, *Structural benchmarks overlook density-of-states errors in machine-learned interatomic potentials*, APL Mach. Learn. **in press** (2026). **DOI**: 10.1063/5.0339982.
49. X. Xu, G. Han, T. Li, J. Hu, G. Qiu, S.-Y. Tseng, G. H. Han, A. Dupuy, A. S. Hall & C. Oses*, *Stabilizing Iodine in Pyrochlore: Toward New Nuclear Waste Forms*, Precis. Chem. **4**(8), 1211–1224 (2026). **DOI**: 10.1021/prechem.5c00467. [PDF]

2025

48. G. Han†, T. Li†, X. Xu, J. Lee, S. Sequeira, A. Ajith & C. Oses*, *The search for high-entropy fuel-cell catalysts using disorder descriptors*, Nano Futures **9**, 045001 (2025). **DOI**: 10.1088/2399-1984/ae19b0. [PDF]
47. C. Oses*, T. Li, X. Xu, G. Han, G. Qiu & J. R. Owens, *Beyond the four core effects: revisiting thermoelectrics with a high-entropy design*, Mater. Horiz. **12**(16), 5946–5956 (2025). **DOI**: 10.1039/D5MH00356C. [PDF]
    - This paper was selected to feature in 2025 Most Popular Articles by the Royal Society of Chemistry: Materials Horizons (2026).
46. G. Qiu, T. Li, X. Xu, Y. Liu, M. Niyogi, K. Cariaga & C. Oses*, *High entropy powering green energy: hydrogen, batteries, electronics, and catalysis*, npj Comput. Mater. **11**, 145 (2025). **DOI**: 10.1038/s41524-025-01594-6. [PDF]

2024

45. T. Gong, G. Qiu, M.-R. He, O. V. Safonova, W.-C. Yang, D. Raciti, C. Oses & A. S. Hall, *Atomic Ordering-Induced Ensemble Variation in Alloys Governs Electrocatalyst On/Off States*, J. Am. Chem. Soc. **147**(1), 510–518 (2024). **DOI**: 10.1021/jacs.4c11753.
44. K. S. Vecchio, S. Curtarolo, K. Kaufmann, T. J. Harrington, C. Oses & C. Toher, *Fermi energy engineering of enhanced plasticity in high-entropy carbides*, Acta Mater. **276**, 120117 (2024). **DOI**: 10.1016/j.actamat.2024.120117. [PDF]
43. M. L. Evans, J. Bergsma, A. Merkys, C. W. Andersen, O. B. Andersson, D. Beltrán, E. Blokhin, T. M. Boland, R. Castañeda Balderas, K. Choudhary, A. Díaz Díaz, R. Domínguez García, H. Eckert, K. Eimre, M. E. Fuentes-Montero, A. M. Krajewski, J. J. Mortensen, J. M. Nápoles-Duarte, J. Pietryga, J. Qi, F. d. J. Trejo Carrillo, A. Vaitkus, J. Yu, A. C. Zettel, P. Baptista de Castro, J. Carlsson, T. F. T. Cerqueira, S. Divilov, H. Hajiyani, F. Hanke, K. Jose, C. Oses, J. Riebesell, J. Schmidt, D. Winston, C. Xie, X. Yang, S. Bonella, S. Botti, S. Curtarolo, C. Draxl, L. E. Fuentes Cobas, A. Hospital, Z.-K. Liu, M. A. L. Marques, N. Marzari, A. J. Morris, S. P. Ong, M. Orozco, K. A. Persson, K. S. Thygesen, C. Wolverton, M. Scheidgen, C. Toher, G. J. Conduit, G. Pizzi, S. Gražulis, G.-M. Rignanese & R. Armiento, *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*, Digit. Discov. **3**, 1509–1533 (2024). **DOI**: 10.1039/D4DD00039K. [PDF]
42. S. Divilov†, H. Eckert†, D. Hicks†, C. Oses†, C. Toher†, R. Friedrich, M. Esters, M. J. Mehl, A. C. Zettel, Y. Lederer, E. Zurek, J.-P. Maria, D. W. Brenner, X. Campilongo, S. Filipović, W. G. Fahrenholtz, C. J. Ryan, C. M. DeSalle, R. J. Crealese, D. E. Wolfe, A. Calzolari & S. Curtarolo, *Disordered enthalpy-entropy descriptor for high-entropy ceramics discovery*, Nature **625**, 66–73 (2024). **DOI**: 10.1038/s41586-023-06786-y. [PDF]

41. A. B. Peters, D. Zhang, S. Chen, C. Ott, C. Oses, S. Curtarolo, I. McCue, T. Pollock & S. E. Prameela, *Materials Design for Hypersonics*, Nat. Commun. **15**, 3328 (2024). **DOI**: 10.1038/s41467-024-46753-3. [PDF]
    - This paper was selected for Editors' Highlight by Springer Nature (2024).

2023

40. D. E. Wolfe, C. M. DeSalle, C. J. Ryan, R. E. Slapikas, R. T. Sweny, R. J. Crealese, P. A. Kolonin, S. P. Stepanoff, A. Haque, S. Divilov, H. Eckert, C. Oses, M. Esters, D. W. Brenner, W. G. Fahrenholtz, J.-P. Maria, C. Toher, E. Zurek & S. Curtarolo, *Influence of Processing on the Microstructural Evolution and Multiscale Hardness in Titanium Carbonitrides (TiCN) Produced via Field Assisted Sintering Technology*, Materialia **27**, 101682 (2023). **DOI**: 10.1016/j.mtla.2023.101682.
39. C. Oses, M. Esters, D. Hicks, S. Divilov, H. Eckert, R. Friedrich, M. J. Mehl, A. Smolyanyuk, X. Campilongo, A. van de Walle, J. Schroers, A. G. Kusne, I. Takeuchi, E. Zurek, M. Buongiorno Nardelli, M. Fornari, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *aflow++: a C++ framework for autonomous materials design*, Comput. Mater. Sci. **217**, 111889 (2023). **DOI**: 10.1016/j.commatsci.2022.111889. [PDF]
    - This paper was selected for Editor's Choice by Elsevier (2022).
38. M. Esters, C. Oses, S. Divilov, H. Eckert, R. Friedrich, D. Hicks, M. J. Mehl, F. Rose, A. Smolyanyuk, A. Calzolari, X. Campilongo, C. Toher & S. Curtarolo, *aflow.org: a web ecosystem of databases, software and tools*, Comput. Mater. Sci. **216**, 111808 (2023). **DOI**: 10.1016/j.commatsci.2022.111808. [PDF]
37. M. Esters†, A. Smolyanyuk†, C. Oses, D. Hicks, S. Divilov, H. Eckert, X. Campilongo, C. Toher & S. Curtarolo, *QH-POCC: taming tiling entropy in thermal expansion calculations of disordered materials*, Acta Mater. **245**, 118594 (2023). **DOI**: 10.1016/j.actamat.2022.118594. [PDF]

2022

36. A. Calzolari, C. Oses, C. Toher, M. Esters, X. Campilongo, S. P. Stepanoff, D. E. Wolfe & S. Curtarolo, *Plasmonic high-entropy carbides*, Nat. Commun. **13**, 5993 (2022). **DOI**: 10.1038/s41467-022-33497-1. [PDF]
35. X. Wang, D. M. Proserpio, C. Oses, C. Toher, S. Curtarolo & E. Zurek, *The Microscopic Diamond Anvil Cell: Stabilization of Superhard, Superconducting Carbon Allotropes at Ambient Pressure*, Angew. Chem. **61**(32), e202205129 (2022). **DOI**: 10.1002/anie.202205129.
34. H. J. Kulik, T. Hammerschmidt, J. Schmidt, S. Botti, M. A. L. Marques, M. Boley, M. Scheffler, M. Todorović, P. Rinke, C. Oses, A. Smolyanyuk, S. Curtarolo, A. Tkatchenko, A. P. Bartók, S. Manzhos, M. Ihara, T. Carrington, J. Behler, O. Isayev, M. Veit, A. Grisafi, J. Nigam, M. Ceriotti, K. T. Schütt, J. Westermayr, M. Gastegger, R. J. Maurer, B. Kalita, K. Burke, R. Nagai, R. Akashi, O. Sugino, J. Hermann, F. Noé, S. Pilati, C. Draxl, M. Kuban, S. Rigamonti, M. Scheidgen, M. Esters, D. Hicks, C. Toher, P. V. Balachandran, I. Tamblyn, S. Whitelam, C. Bellinger & L. M. Ghiringhelli, *Roadmap on Machine Learning in Electronic Structure*, Electron. Struct. **4**(2), 023004 (2022). **DOI**: 10.1088/2516-1075/ac572f. [PDF]
33. A. G. Kusne, A. McDannald, B. DeCost, C. Oses, C. Toher, S. Curtarolo, A. Mehta & I. Takeuchi, *Physics in the Machine: Integrating Physical Knowledge in Autonomous Phase-Mapping*, Front. Phys. **10**, 815863 (2022). **DOI**: 10.3389/fphy.2022.815863. [PDF]
32. C. Toher, C. Oses, M. Esters, D. Hicks, G. N. Kotsonis, C. M. Rost, D. W. Brenner, J.-P. Maria & S. Curtarolo, *High-entropy ceramics: Propelling applications through disorder*, MRS Bull. **47**, 194–202 (2022). **DOI**: 10.1557/s43577-022-00281-x.

2021

31. M. Esters, C. Oses, D. Hicks, M. J. Mehl, M. Jahnátek, M. D. Hossain, J.-P. Maria, D. W. Brenner, C. Toher & S. Curtarolo, *Settling the matter of the role of vibrations in the stability of high-entropy carbides*, Nat. Commun. **12**, 5747 (2021). **DOI**: 10.1038/s41467-021-25979-5. [PDF]
    - This paper was selected for Editors' Highlight by Springer Nature (2021).
30. M. D. Hossain, T. Borman, C. Oses, M. Esters, C. Toher, L. Feng, A. Kumar, W. G. Fahrenholtz, S. Curtarolo, D. W. Brenner, J. M. LeBeau & J.-P. Maria, *Entropy Landscaping of High-Entropy Carbides*, Adv. Mater. **33**(42), 2102904 (2021). **DOI**: 10.1002/adma.202102904.
29. C. W. Andersen†, R. Armiento†, E. Blokhin†, G. J. Conduit†, S. Dwaraknath†, M. L. Evans†, Á. Fekete†, A. Gopakumar†, S. Gražulis†, A. Merkys†, F. Mohamed†, C. Oses†, G. Pizzi†, G.-M. Rignanese†, M. Scheidgen†, L. Talirz†, C. Toher†, D. Winston†, R. Aversa, K. Choudhary, P. Colinet, S. Curtarolo, D. Di Stefano, C. Draxl, S. Er, M. Esters, M. Fornari, M. Giantomassi, M. Govoni, G. Hautier, V. Hegde, M. K. Horton, P. Huck, G. Huhs, J. Hummelshøj, A. Kariryaa, B. Kozinsky, S. Kumbhar, M. Liu, N. Marzari, A. J. Morris, A. Mostofi, K. A. Persson, G. Petretto, T. Purcell, F. Ricci, F. Rose, M. Scheffler, D. Speckhard, M. Uhrin, A. Vaitkus, P. Villars, D. Waroquiers, C. Wolverton, M. Wu & X. Yang, *OPTIMADE: an API for exchanging materials data*, Sci. Data **8**, 217 (2021). **DOI**: 10.1038/s41597-021-00974-z. [PDF]
28. R. Friedrich, M. Esters, C. Oses, S. Ki, M. J. Brenner, D. Hicks, M. J. Mehl, C. Toher & S. Curtarolo, *Automated coordination corrected enthalpies with AFLOW-CCE*, Phys. Rev. Mater. **5**, 043803 (2021). **DOI**: 10.1103/PhysRevMaterials.5.043803.
27. D. Hicks, M. J. Mehl, M. Esters, C. Oses, O. Levy, G. L. W. Hart, C. Toher & S. Curtarolo, *The AFLOW Library of Crystallographic Prototypes: Part 3*, Comput. Mater. Sci. **199**, 110450 (2021). **DOI**: 10.1016/j.commatsci.2021.110450.
26. M. J. Mehl, M. Ronquillo, D. Hicks, M. Esters, C. Oses, R. Friedrich, A. Smolyanyuk, E. Gossett, D. Finkenstadt & S. Curtarolo, *Tin-pest problem as a test of density functionals using high-throughput calculations*, Phys. Rev. Mater. **5**, 083608 (2021). **DOI**: 10.1103/PhysRevMaterials.5.083608.
25. M. D. Hossain†, T. Borman†, A. Kumar, X. Chen, A. Khosravani, S. R. Kalidindi, E. A. Paisley, M. Esters, C. Oses, C. Toher, S. Curtarolo, J. M. LeBeau, D. W. Brenner & J.-P. Maria, *Carbon Stoichiometry and Mechanical Properties of High Entropy Carbides*, Acta Mater. **215**, 117051 (2021). **DOI**: 10.1016/j.actamat.2021.117051.

2020

24. A. G. Kusne†, H. Yu†, C. Wu, H. Zhang, J. Hattrick-Simpers, B. DeCost, S. Sarker, C. Oses, C. Toher, S. Curtarolo, A. V. Davydov, R. Agarwal, L. A. Bendersky, M. Li, A. Mehta & I. Takeuchi, *On-the-fly Closed-loop Autonomous Materials Discovery via Bayesian Active Learning*, Nat. Commun. **11**, 5966 (2020). **DOI**: 10.1038/s41467-020-19597-w. [PDF]
23. K. Kaufmann, D. Maryanovsky, W. M. Mellor, C. Zhu, A. S. Rosengarten, T. J. Harrington, C. Oses, C. Toher, S. Curtarolo & K. S. Vecchio, *Discovery of novel high-entropy ceramics via machine learning*, npj Comput. Mater. **6**, 42 (2020). **DOI**: 10.1038/s41524-020-0317-6. [PDF]
22. C. Oses, C. Toher & S. Curtarolo, *High-entropy ceramics*, Nat. Rev. Mater. **5**, 295–309 (2020). **DOI**: 10.1038/s41578-019-0170-8.
    - This paper was highlighted as a "hot paper" by Web of Science (Clarivate Analytics) (2021).

2019

21. D. C. Ford, D. Hicks, C. Oses, C. Toher & S. Curtarolo, *Metallic glasses for biodegradable implants*, Acta Mater. **176**, 297–305 (2019). **DOI**: 10.1016/j.actamat.2019.07.008.
20. P. Avery, X. Wang, C. Oses, E. Gossett, D. M. Proserpio, C. Toher, S. Curtarolo & E. Zurek, *Predicting Superhard Materials via a Machine Learning Informed Evolutionary Structure Search*, npj Comput. Mater. **5**, 89 (2019). **DOI**: 10.1038/s41524-019-0226-8. [PDF]
19. C. Toher, C. Oses, D. Hicks & S. Curtarolo, *Unavoidable disorder and entropy in multi-component systems*, npj Comput. Mater. **5**, 69 (2019). **DOI**: 10.1038/s41524-019-0206-z. [PDF]
18. R. Friedrich, D. Usanmaz, C. Oses, A. R. Supka, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *Coordination corrected ab initio formation enthalpies*, npj Comput. Mater. **5**, 59 (2019). **DOI**: 10.1038/s41524-019-0192-1. [PDF]
17. P. Nath, D. Usanmaz, D. Hicks, C. Oses, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *AFLOW-QHA3P: Robust and automated method to compute thermodynamic properties of solids*, Phys. Rev. Mater. **3**, 073801 (2019). **DOI**: 10.1103/PhysRevMaterials.3.073801.

2018

16. C. Oses, E. Gossett, D. Hicks, F. Rose, M. J. Mehl, E. Perim, I. Takeuchi, S. Sanvito, M. Scheffler, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *AFLOW-CHULL: Cloud-oriented platform for autonomous phase stability analysis*, J. Chem. Inf. Model. **58**(12), 2477–2490 (2018). **DOI**: 10.1021/acs.jcim.8b00393.
15. C. Oses, C. Toher & S. Curtarolo, *Data-driven design of inorganic materials with the Automatic Flow Framework for Materials Discovery*, MRS Bull. **43**(9), 670–675 (2018). **DOI**: 10.1557/mrs.2018.207.
14. P. Sarker†, T. J. Harrington†, C. Toher, C. Oses, M. Samiee, J.-P. Maria, D. W. Brenner, K. S. Vecchio & S. Curtarolo, *High-entropy high-hardness metal carbides discovered by entropy descriptors*, Nat. Commun. **9**, 4980 (2018). **DOI**: 10.1038/s41467-018-07160-7. [PDF]
13. V. Stanev, C. Oses, A. G. Kusne, E. Rodriguez, J. Paglione, S. Curtarolo & I. Takeuchi, *Machine learning modeling of superconducting critical temperature*, npj Comput. Mater. **4**, 29 (2018). **DOI**: 10.1038/s41524-018-0085-8. [PDF]
12. E. Gossett, C. Toher, C. Oses, O. Isayev, F. Legrain, F. Rose, E. Zurek, J. Carrete, N. Mingo, A. Tropsha & S. Curtarolo, *AFLOW-ML: A RESTful API for machine-learning prediction of materials properties*, Comput. Mater. Sci. **152**, 134–145 (2018). **DOI**: 10.1016/j.commatsci.2018.03.075.
    - This paper was selected for Editor's Choice by Elsevier (2018).
11. D. Hicks, C. Oses, E. Gossett, G. Gomez, R. H. Taylor, C. Toher, M. J. Mehl, O. Levy & S. Curtarolo, *AFLOW-SYM: platform for the complete, automatic and self-consistent symmetry analysis of crystals*, Acta Cryst. A **74**, 184–203 (2018). **DOI**: 10.1107/S2053273318003066.

2017

10. A. Hever, C. Oses, S. Curtarolo, O. Levy & A. Natan, *The structure and composition statistics of 6A binary and ternary structures*, Inorg. Chem. **57**(2), 653–667 (2017). **DOI**: 10.1021/acs.inorgchem.7b02462.
9. F. Rose, C. Toher, E. Gossett, C. Oses, M. Buongiorno Nardelli, M. Fornari & S. Curtarolo, *AFLUX: The LUX materials search API for the AFLOW data repositories*, Comput. Mater. Sci. **137**, 362–370 (2017). **DOI**: 10.1016/j.commatsci.2017.04.036.
    - This paper was selected for Editor's Choice by Elsevier (2017).
8. O. Isayev†, C. Oses†, C. Toher, E. Gossett, S. Curtarolo & A. Tropsha, *Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals*, Nat. Commun. **8**, 15679 (2017). **DOI**: 10.1038/ncomms15679. [PDF]
7. C. Toher, C. Oses, J. J. Plata, D. Hicks, F. Rose, O. Levy, M. de Jong, M. Asta, M. Fornari, M. Buongiorno Nardelli & S. Curtarolo, *Combining the AFLOW GIBBS and elastic libraries to efficiently and robustly screen thermomechanical properties of solids*, Phys. Rev. Mater. **1**, 015401 (2017). **DOI**: 10.1103/PhysRevMaterials.1.015401.
6. C. Nyshadham, C. Oses, J. E. Hansen, I. Takeuchi, S. Curtarolo & G. L. W. Hart, *A Computational High-Throughput Search for New Ternary Superalloys*, Acta Mater. **122**, 438–447 (2017). **DOI**: 10.1016/j.actamat.2016.09.017.
5. S. Sanvito, C. Oses, J. Xue, A. Tiwari, M. Žic, T. Archer, P. Tozman, M. Venkatesan, J. M. D. Coey & S. Curtarolo, *Accelerated Discovery of New Magnets in the Heusler Alloy Family*, Sci. Adv. **3**(4), e1602241 (2017). **DOI**: 10.1126/sciadv.1602241. [PDF]

2016

4. A. van Roekeghem, J. Carrete, C. Oses, S. Curtarolo & N. Mingo, *High-Throughput Computation of Thermal Conductivity of High-Temperature Solid Phases: The Case of Oxide and Fluoride Perovskites*, Phys. Rev. X **6**(4), 041061 (2016). **DOI**: 10.1103/PhysRevX.6.041061. [PDF]

3. K. Yang, C. Oses & S. Curtarolo, *Modeling Off-Stoichiometry Materials with a High-Throughput Ab-Initio Approach*, Chem. Mater. **28**(18), 6484–6492 (2016). **DOI**: 10.1021/acs.chemmater.6b01449.

2015

2. C. E. Calderon, J. J. Plata, C. Toher, C. Oses, O. Levy, M. Fornari, A. Natan, M. J. Mehl, G. L. W. Hart, M. Buongiorno Nardelli & S. Curtarolo, *The AFLOW Standard for High-Throughput Materials Science Calculations*, Comput. Mater. Sci. **108A**, 233–238 (2015). **DOI**: 10.1016/j.commatsci.2015.07.019.
   - This paper was selected for Editor's Choice by Elsevier (2015).
1. O. Isayev, D. Fourches, E. N. Muratov, C. Oses, K. M. Rasch, A. Tropsha & S. Curtarolo, *Materials Cartography: Representing and Mining Materials Space Using Structural and Electronic Fingerprints*, Chem. Mater. **27**(3), 735–743 (2015). **DOI**: 10.1021/cm503507h. [PDF]
   - This paper was one of the top 10 most highly downloaded papers for the month of January 2015 by the American Chemical Society (2015).
   - This paper was selected for Editors' Choice by the American Chemical Society (2015).

PDF links are provided for open-access articles and for the author's own version where the publisher permits it. For a copy of any other article, please email me.

Book Publications

2019

3. C. Toher, C. Oses & S. Curtarolo, *Automated computation of materials properties*, Materials Informatics: Methods, Tools and Applications, Ch. 7. **DOI**: 10.1002/9783527802265.ch7.

2018

2. S. Sanvito, M. Žic, J. Nelson, T. Archer, C. Oses & S. Curtarolo, *Machine learning and high-throughput approaches to magnetism*, Handbook of Materials Modeling. Volume 2 Applications: Current and Emerging Materials. **DOI**: 10.1007/978-3-319-50257-1_108-1.
1. C. Toher, C. Oses, D. Hicks, E. Gossett, F. Rose, P. Nath, D. Usanmaz, D. C. Ford, E. Perim, C. E. Calderon, J. J. Plata, Y. Lederer, M. Jahnátek, W. Setyawan, S. Wang, J. Xue, K. M. Rasch, R. V. Chepulskii, R. H. Taylor, G. Gomez, H. Shi, A. R. Supka, R. Al Rahal Al Orabi, P. Gopal, F. T. Cerasoli, L. Liyanage, H. Wang, I. Siloi, L. A. Agapito, C. Nyshadham, G. L. W. Hart, J. Carrete, F. Legrain, N. Mingo, E. Zurek, O. Isayev, A. Tropsha, S. Sanvito, R. M. Hanson, I. Takeuchi, M. J. Mehl, A. N. Kolmogorov, K. Yang, P. D'Amico, A. Calzolari, M. Costa, R. De Gennaro, M. Buongiorno Nardelli, M. Fornari, O. Levy & S. Curtarolo, *The AFLOW Fleet for Materials Discovery*, Handbook of Materials Modeling. Volume 1 Methods: Theory and Modeling. **DOI**: 10.1007/978-3-319-42913-7_63-2.

Talks/Presentations

2026

45. *Powering Tomorrow with AI-Driven High-Entropy Design*. **Invited talk** for the Symposium on "High-Entropy Ceramics" at the CIMTEC International Conferences on Modern Materials and Technologies 2026; 16th International Ceramics Congress, Perugia, Italy (Jun 21, 2026).
44. *Powering Tomorrow with AI-Driven High-Entropy Design*. **Invited talk** for the Session on "Computational Materials" at the GE Vernova's Material Frontier Symposium: Powering the Future, Niskayuna, NY (Jun 03, 2026).
43. *Powering Tomorrow with AI-Driven High-Entropy Design*. **Invited talk** for the Symposium on "Engineering Complexity—High Entropy Ceramics as a Design Platform for Next-Generation Materials" at the MRS 2026 Spring Meeting & Exhibit, Honolulu, HI (May 01, 2026).
42. *The Intersection of Energy, Entropy, and Exploration: Data-Driven Discovery of High-Entropy Materials*. **Invited talk** for the Session on "Data-Driven Thin Film Design: High-Throughput Experimentation, Simulation, and Machine Learning" at the International Conference on Metallurgical Coatings and Thin Films (ICMCTF) 2026, San Diego, CA (Apr 20, 2026).
41. *Powering Tomorrow with AI-Driven High-Entropy Design*. **Invited talk** for the Material Sciences Strategy Offsite for the Johns Hopkins University Applied Physics Laboratory Section R1A, Washington, DC (Mar 26, 2026).
40. *Powering Tomorrow with AI-Driven High-Entropy Design*. **Invited seminar** for the Departmental Seminar Series at the Department of Materials Science and Engineering at the University of Maryland, College Park, MD (Mar 25, 2026).
39. *From Data to Discovery: Active Learning Unlocks Complex Ceramic Design Spaces*. **Invited talk** for the Symposium on "Advances in Ceramic Materials and Processing" at the TMS 2026 Annual Meeting & Exhibition, San Diego, CA (Mar 16, 2026).

2025

38. *From Data to Discovery: Active Learning Unlocks Complex Ceramic Design Spaces*. **Invited talk** for the Mini Symposium on "Computational Thermodynamics: Energy and Energy Landscape" at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, University Park, PA (Nov 02, 2025).
37. *AI Materials Discovery for Nuclear Waste and Spent Fuel Immobilization*. **Invited talk** for the 2025 Annual Fission Workshop of the Advanced Research Projects Agency-Energy (ARPA-E), Arlington, VA (Oct 02, 2025).

36. *High-Entropy Alloys and Halides: Expanding the Energy-Materials Space*. **Invited talk** for the Symposium on "Advances in Refractory High Entropy Alloys and Ceramics" at the MS&T25 Technical Meeting and Exhibition, Columbus, OH (Sep 30, 2025).
35. *Accelerating Energy Solutions with High-Entropy Materials: Leveraging Disorder, Computation, and AI*. **Invited talk** for The Advanced Materials Show at MS&T25 Technical Meeting and Exhibition, Columbus, OH (Sep 30, 2025).
34. *High-Entropy Oxides and Halides: Expanding the Energy-Materials Space*. **Invited talk** for the Artificial Intelligence for Materials Science (AIMS) Workshop at the National Institute of Standards and Technology (NIST), Rockville, MD (Jun 10, 2025).
    - *Artificial Intelligence for Materials Science (AIMS) Workshop* recording: https://www.nist.gov/news-events/events/2025/07/artificial-intelligence-materials-science-aims-workshop
33. *High-Entropy Halides: Expanding the Energy-Materials Space*. **Invited talk** for the Symposium on "High Entropy and Complex Structure for Electrocatalysis and Other Applications" at the ACS Spring 2025 Meeting & Expo, San Diego, CA (Mar 25, 2025).
32. *High-Entropy Halides: Expanding the Energy-Materials Space*. **Invited talk** for the Symposium on "Advances in Ceramic Materials and Processing" at the TMS 2025 Annual Meeting & Exhibition, Las Vegas, NV (Mar 24, 2025).

2024

31. *Metal Iodide Materials for Energy Applications*. **Invited talk** for the Symposium on "Understanding High Entropy Materials Via Data-Science and Computational Approaches" at the MS&T24 Technical Meeting and Exhibition, Pittsburgh, PA (Oct 09, 2024).
30. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the Departmental Seminar Series at the Department of Materials Science and Engineering at the University of Connecticut, Storrs, CT (Sep 27, 2024).
29. *Disorder by design: Applications and modeling of high-entropy ceramics*. **Invited talk** for the Symposium on "Advancing Ab-Initio Force Fields with Machine-Learning for Energy Materials" at the International Conference on Computational & Experimental Engineering and Sciences (ICCES 2024), Singapore (Aug 04, 2024).
28. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the Departmental Seminar Series at the Department of Chemical and Nano Engineering at the University of California, San Diego, San Diego, CA (May 01, 2024).
27. *Disorder by design: Applications and modeling of high-entropy ceramics*. **Invited seminar** for the Departmental Seminar Series at the University of Michigan, Ann Arbor, MI (Mar 04, 2024).
26. *Bayesian Optimization of the PhD (and beyond)*. **Keynote** for the Fifth Annual Research Summit for the University Center of Exemplary Mentoring at Duke University, Durham, NC (Feb 29, 2024).

2023

25. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the Materials Science and Engineering Fall Seminar Series at the Rensselaer Polytechnic Institute, Troy, NY (Oct 18, 2023).
24. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the Physics Department Colloquium at Georgetown University, Washington, DC (Sep 26, 2023).
23. *Computational Materials Science*. **Invited seminar** for the PARADIM Summer School, Johns Hopkins University (Aug 02, 2023).
22. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the OneChemistry Symposium, Johns Hopkins University (Apr 18, 2023).
21. *Disorder by design: Applications and modeling of high-entropy ceramics*. **Invited seminar** for the Johns Hopkins University Applied Physics Laboratory, Baltimore, Maryland (Mar 09, 2023).
20. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the Computational Spintronics Group, Trinity College Dublin, Ireland (Feb 17, 2023).
19. *Success Stories in Computationally–Driven Materials Discovery*. **Invited seminar** for the Physics Department Colloquium at Johns Hopkins University, Baltimore, Maryland (Feb 15, 2023).

2022

18. *Disorder by design: Applications and modeling of high-entropy ceramics*. **Invited seminar** for the Hopkins Extreme Materials Institute at Johns Hopkins University, Baltimore, Maryland (Nov 01, 2022).
17. *Formation Descriptors for High-Entropy High-Hardness Metal Carbides*. **Invited talk** for the Machine Learning in Ceramics and Glasses Webinar at the Institute of Materials, Minerals and Mining (IOM3), London, UK (Mar 29, 2022).
16. *High-entropy ceramics*. **Invited seminar** for the Department of Materials Science and Engineering at Michigan Technological University, Houghton, Michigan (Mar 17, 2022).
15. *High-entropy ceramics*. **Invited seminar** for the Department of Materials Science and Engineering at the University of California, Irvine, Irvine, CA (Mar 04, 2022).
14. *Disorder by design: Applications and modeling of high-entropy ceramics*. **Invited seminar** for the Department of Materials Science and Engineering at Texas A&M University, College Station, Texas (Feb 10, 2022).
13. *High-entropy ceramics*. **Invited seminar** for the Department of Physics at the University of Alabama at Birmingham, Birmingham, Alabama (Feb 04, 2022).
12. *High-entropy ceramics*. **Invited seminar** for the Department of Mechanical Engineering at Rowan University, Glassboro, NJ (Jan 27, 2022).
11. *High-entropy ceramics*. **Invited seminar** for the Department of Materials Science and Engineering at Johns Hopkins University, Baltimore, MD (Jan 04, 2022).

2021

10. *Data for Materials Development Platforms*. **Invited seminar** for the aiM Program Boot Camp and Orientation at Duke University, Durham, North Carolina (Aug 19, 2021).
    - *Data for Materials Development Platforms* recording: https://youtu.be/wLegemRIMpk
9. *High-entropy ceramics*. **Invited seminar** for the Department of Mechanical Engineering at Texas A&M University, College Station, Texas (Feb 24, 2021).
8. *High-entropy ceramics*. **Invited seminar** for the Lecture Series in Materials Science & Engineering at the North Carolina State University, Raleigh, North Carolina (Jan 22, 2021).

2020

7. *Entropy and ceramics: A valuable partnership*. **Invited seminar** for the Department of Materials and Interfaces at the Weizmann Institute of Science, Rehovot, Israel (Feb 06, 2020).
6. *Entropy and ceramics: A valuable partnership*. **Invited seminar** for the Sackler Center for Computational Molecular and Materials Science at Tel Aviv University, Tel Aviv, Israel (Feb 05, 2020).
5. *Entropy and ceramics: A valuable partnership*. **Invited seminar** for the Department of Materials Engineering at Ben-Gurion University of the Negev, Beer Sheva, Israel (Jan 29, 2020).

2019

4. *Going Off-Stoichiometry: Challenging Traditional Materials Discovery*. **Invited seminar** for the Center for Computational Materials Science at the Naval Research Laboratory, Washington, DC (Jan 09, 2019).

2018

3. *Advancements in Materials Informatics with AFLOW*. **Invited seminar** for the Theory Department at the Fritz-Haber-Institut der Max-Planck-Gesellschaft, Berlin, Germany (Jan 18, 2018).
2. *Advancements in Materials Informatics with AFLOW*. **Invited seminar** for the Physics Department at the Humboldt University of Berlin, Berlin, Germany (Jan 16, 2018).

2016

1. *Materials Cartography: Representing and Mining Materials Space using Structural and Electronic Fingerprints*. **Invited seminar** for the Condensed Matter Physics Seminar Series at Brigham Young University, Provo, Utah (Feb 18, 2016).

Software and Datasets

Prior development, Duke University (2014–2022)

aflow++ (2014–2022) · Lead developer · C++ framework for autonomous materials design and high-throughput ab initio workflows · with the AFLOW Consortium (S. Curtarolo, PI, Duke University) · **DOI**: 10.1016/j.commatsci.2022.111889 · aflow.org · Also developed the electronic band-gap and Bader charge analysis modules; `POCC`, `CHULL` and `APL` are listed separately below · Continuing development of a Johns Hopkins-specific branch (2022–present).

AFLOW-POCC (2014–2022) · Lead developer · partial-occupation and disorder module for modeling off-stoichiometric and high-entropy materials · with the AFLOW Consortium · **DOI**: 10.1021/acs.chemmater.6b01449 & 10.1016/j.actamat.2022.118594.

AFLOW-CHULL (2014–2022) · Lead developer · cloud-oriented platform for autonomous thermodynamic phase-stability analysis · with the AFLOW Consortium · **DOI**: 10.1021/acs.jcim.8b00393 · aflow.org/aflow-chull.

AFLOW-APL (2014–2019) · Core developer · automatic phonon library for vibrational and thermal properties · with the AFLOW Consortium · **DOI**: 10.1038/s41467-021-25979-5.

aflow.org (2014–2022) · Core developer · web ecosystem of materials databases, software and tools · with the AFLOW Consortium · **DOI**: 10.1016/j.commatsci.2022.111808 · aflow.org.

AFLOW-SYM (2014–2019) · Contributing developer · complete, automatic and self-consistent symmetry analysis of crystals · with the AFLOW Consortium · **DOI**: 10.1107/S2053273318003066 · aflow.org/aflow-online.

AFLOW-CCE (2021) · Contributing developer · automated coordination-corrected formation enthalpies · with the AFLOW Consortium · **DOI**: 10.1103/PhysRevMaterials.5.043803 · aflow.org/aflow-online.

AFLOW-ML (2018) · Contributing developer · RESTful API for machine-learning prediction of materials properties · with the AFLOW Consortium · **DOI**: 10.1016/j.commatsci.2018.03.075 · aflow.org/aflow-ml.

AFLUX (2017) · Contributing developer · search API for the AFLOW data repositories · with the AFLOW Consortium · **DOI**: 10.1016/j.commatsci.2017.04.036 · aflow.org/API/aflux.

OPTIMADE (2021–2024) · Contributor · community API standard for exchanging materials data across databases · with the OPTIMADE Consortium · **DOI**: 10.1038/s41597-021-00974-z & 10.1039/D4DD00039K · optimade.org.

Teaching Experience

Instructor | Spring 2023–present | EN.500.113: *Gateway Computing: Python*, Johns Hopkins University

- Required first-year coding course for all Whiting School undergraduates; one section in Spring 2023 and two sections every spring since, 20 to 32 students per year
- Project-based flipped classroom with in-class team problems solved at the whiteboard; authored one of the class-wide projects (string manipulation applied to order and disorder in materials)
- Course evaluations (seven sections, 91 respondents, response rates 86 to 100%): teaching effectiveness 4.25, intellectual challenge 4.34 of 5 (school means 4.27 and 4.28); Spring 2023 section 4.69 for teaching effectiveness

Instructor | Fall 2023–present | EN.510.466 / EN.510.666: *Materials Modeling (formerly Introduction to Computational Materials Modeling)*, Johns Hopkins University

- Cross-listed undergraduate and graduate course; enrollment 15 to 17 per year, mostly graduate students, with a waiting list in 2026
- Seven programming assignments with LaTeX reports in GitHub, no written examinations, final group project; closes with autonomous experimentation on the LEGOLAS kit
- Course evaluations (three years, 44 respondents, response rates 80 to 94%): quality 4.30, teaching effectiveness 4.30, intellectual challenge 4.62 of 5 (school mean for challenge about 4.26)

Research Advisor | Spring 2023–present | EN.510.501 / EN.510.502 / EN.510.809: *Undergraduate Research, Research in Materials Science, and Graduate Summer Research*, Johns Hopkins University

Guest Lecturer | Feb 05, 2026 | EN.510.603: *Phase Transformations (T. Curk)*, Johns Hopkins University

Guest Lecturer | Mar 28, 2023 | EN.510.422 / EN.510.622: *Micro and Nano Structured Materials and Devices (A. S. Hall)*, Johns Hopkins University

Co-Instructor | Spring 2021 | ME 555: *Applications of Artificial Intelligence in Materials*, Duke University Department of Mechanical Engineering and Materials Science

Teaching Assistant | Spring 2020 | ME 555: *Computational Materials Science by Examples and Applications*, Duke University Department of Mechanical Engineering and Materials Science

Teaching Assistant | Fall 2014–Spring 2015 | ME 221: *Structure and Properties of Solids*, Duke University Department of Mechanical Engineering and Materials Science

- Best Teaching Assistant Award, Aug 14, 2015

Conference and Workshop Organization † contributed equally

2026

31. *AI-Enabled Energy Systems: Technologies, Intelligence, and Security Conference*. **Organizer** at Johns Hopkins University, Baltimore, Maryland (Oct 28, 2026).
30. *Laboratory to Launchpad: Bringing Your Energy IP to Life*. **Organizer** at Hopkins Bloomberg Center, Washington, DC (Jun 11, 2026). **Co-Organizers**: A. G. Bregman†, S. D. Cohen & J. Erlebacher.
    - website: https://engineering.jhu.edu/materials/laboratory-to-launchpad/
29. *Navigating Early Careers in Academia*. **Panelist** for the Materials Graduate Society at Johns Hopkins University, Baltimore, Maryland (Apr 15, 2026).
28. *Quantum Computing in Practice: Hands On with Quantinuum*. **Organizer** at Johns Hopkins University, Baltimore, Maryland (Feb 25–26, 2026). **Co-Organizer**: G. D. Quiroz.
    - website: https://entropy4energy.ai/workshops

2025

27. *Mini Symposium on "Computational Thermodynamics: Energy and Energy Landscape"*. **Co-Organizer and Presenter** at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, University Park, PA (Oct 31–Nov 2, 2025). **Co-Organizers**: Z.-K. Liu, J. Deng, T. R. Sinno & R. Wentzcovitch.
    - website: https://siamnnp2025.sciencesconf.org/
26. *Hopkins Engineering Sampler Seminar (HESS)*. **Panelist** for the Life Design – First Year Experience (Whiting School of Engineering) at Johns Hopkins University, Baltimore, Maryland (Sep 22, 2025).
25. *Johns Hopkins Data Science and AI Institute Fall 2025 Symposium: AI in Daily Life*. **Organizer** for the Fall 2025 Symposium at the Johns Hopkins Data Science and AI Institute, Baltimore, MD (Sep 16, 2025). **Co-Organizers**: D. J. H. Mathews†, S. Vedula†, C. Douville†, T. Shu†, K. Vaught†, B. Ranno, S. Pagano & S. McGhee.
    - website: https://ai.jhu.edu/event/fall-2025-symposium/

2024

24. *Data-Driven Materials Modeling Workshop*. **Organizer and Presenter** at Johns Hopkins University, Baltimore, Maryland (May 29–31, 2024). **Co-Organizers**: B. Bukowski & T. Curk.
    - *Data-Driven Thermodynamic Modeling for Materials Discovery* recording: https://youtu.be/kZj3zQkBAKg
    - website: https://entropy4energy.ai/workshops
23. *Faculty Panel*. **Panelist** for the Engineering Admitted Student Visit Day at Johns Hopkins University, Baltimore, Maryland (Mar 26, 2024).
22. *Focus Session: Computational Design, Understanding and Discovery of Novel Materials*. **Session Chair** for the March Meeting of the American Physical Society, Minneapolis, Minnesota (Mar 3–8, 2024). **Co-Chairs**: E. Jankowski, R. Sundararaman & D. Usanmaz.
    - website: https://meetings-archive.aps.org/mar/2024/g60/
21. *Foundations to Futures: Materials Data and AI*. **Conference Co-Chair** at the Materials Research Data Alliance (MaRDA) 2024 Annual Meeting, Baltimore, Maryland (Feb 20–22, 2024). **Co-Chairs**: D. Audus & F. Sen.
    - website: https://www.marda-alliance.org/2024-annual-meeting/
20. *AI, Data Science — Developing the Role for Sustainable Energy in Hopkins' Expansion and Vision*. **Session Co-Chair** at the ROSEI 2024 Summit, Baltimore, Maryland (Jan 17, 2024). **Co-Chair**: P. Clancy.
    - website: https://energyinstitute.jhu.edu/pec-events/rosei-summit-2024/

2023

19. *Hopkins Engineering Sampler Seminar (HESS)*. **Panelist** for the Life Design – First Year Experience (Whiting School of Engineering) at Johns Hopkins University, Baltimore, Maryland (Sep 19, 2023).
18. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST/MRS, College Park, Maryland (Aug 10, 2023). **Co-Presenter**: C. Toher.
    - website: https://umd-mse.catalog.instructure.com/courses/mlmr-2023
17. *Roundtable Discussion on Gateway Science/Computing Courses*. **Panelist** for the Provost's DELTA Teaching Forum at Johns Hopkins University, Baltimore, Maryland (May 01, 2023).

2022

16. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Organizer and Presenter** at Johns Hopkins University, Baltimore, Maryland (Sep 21, 2022). **Co-Organizers**: D. Hicks, H. Eckert & S. Divilov.
    - *Introduction and AFLOW-ML: Machine Learning* recording: https://youtu.be/Xj5BGuFC9ew
    - website: https://aflow.org/aflow-school/#20220921
15. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST/MRS, College Park, Maryland (Aug 11, 2022). **Co-Presenters**: C. Toher & D. Hicks.
    - website: https://aflow.org/aflow-school/#20220811
14. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at the East African Institute for Fundamental Research, University of Rwanda, Kigali, Rwanda (Feb 21–24, 2022). **Co-Organizers**: C. Toher, M. Esters & F. Rose.
    - website: https://aflow.org/aflow-school/#20220221

2021

13. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at the Technische Universität (TU) Dresden and Helmholtz-Zentrum Dresden-Rossendorf (Sep 6–10, 2021). **Co-Organizers**: C. Toher, D. Hicks, M. Esters, A. Smolyanyuk & R. Friedrich.
    - *Introduction to Density Functional Theory and VASP* recording: https://youtu.be/_RsQH3TY7kI
    - *AFLOW-CHULL: Thermodynamics* recording: https://youtu.be/zcY7gTZIB-Y
    - *AFLOW-POCC: Disorder* recording: https://youtu.be/lcDSYiF4AS4
    - website: https://aflow.org/aflow-school/#20210906
12. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at the University of Virginia, Charlottesville, Virginia (Aug 17, 2021). **Co-Organizers**: C. Toher, D. Hicks & M. Esters.
    - *AFLOW-CHULL and AFLOW-CCE: Thermodynamics* recording: https://youtu.be/cLhOcN1sQ7M
    - website: https://aflow.org/aflow-school/#20210817
11. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST, College Park, Maryland (Jul 29, 2021). **Co-Presenters**: C. Toher, D. Hicks & M. Esters.
    - *AFLOW-ML: Machine Learning* recording: https://youtu.be/uFQ-lyTaxCc
    - website: https://aflow.org/aflow-school/#20210729
10. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at Texas A&M University, College Station, Texas (Jul 12–15, 2021). **Co-Organizers**: C. Toher, D. Hicks, M. Esters & A. Smolyanyuk.
    - *Introduction to Density Functional Theory and VASP* recording: https://youtu.be/KXnJGdVgosA
    - *AFLOW-CHULL and AFLOW-CCE: Thermodynamics* recording: https://youtu.be/ElaniAcrbhU
    - *AFLOW-POCC: Disorder* recording: https://youtu.be/D_cfHIlpBiA
    - website: https://aflow.org/aflow-school/#20210712
9. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Session Chair** for the Virtual Spring Meeting of the Materials Research Society (Apr 17, 2021). **Co-Chairs**: C. Toher, D. Hicks & M. Esters.
    - website: https://mrs2021.cd.pathable.com/meetings/virtual/ZwzoadbK59GonYZuw

2020

8. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Materials 4.0 Summer School 2020 at the Dresden Center for Computational Materials Science (DCMS), Technische Universität (TU) Dresden (Aug 18, 2020). **Co-Presenters**: C. Toher, D. Hicks, M. Esters & R. Friedrich.
    - *AFLOW-CHULL: Thermodynamics* recording: https://youtu.be/ncm356YNBVc
    - website: https://aflow.org/aflow-school/#20200818
7. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Machine Learning Microscopy Data of the University of Maryland/NIST, College Park, Maryland (Jul 23, 2020). **Co-Presenters**: C. Toher, D. Hicks & M. Esters.
    - *AFLOW-ML: Machine Learning* recording: https://youtu.be/x2qeBtOXues
    - website: https://www.nanocenter.umd.edu/events/mlmr-2020/
6. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at Texas A&M University, College Station, Texas (Jun 16–18, 2020). **Co-Organizers**: C. Toher, D. Hicks, M. Esters & R. Friedrich.
    - *Introduction to Density Functional Theory and VASP* recording: https://youtu.be/ChySAfo2w7g
    - *AFLOW-CHULL: Thermodynamics* recording: https://youtu.be/9Sa8D4inJ5w
    - *AFLOW-POCC: Disorder* recording: https://youtu.be/xr-mU-1ShQQ
    - website: https://aflow.org/aflow-school/#20200616

2019

5. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Autonomous Materials Research of the University of Maryland/NIST, College Park, Maryland (Aug 05, 2019). **Co-Presenters**: C. Toher, D. Hicks & M. Esters.
    - website: https://www.nanocenter.umd.edu/events/mlmr/program/
4. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at the University of Pennsylvania, Philadelphia, Pennsylvania (May 03, 2019). **Co-Organizers**: C. Toher, D. Hicks, E. Gossett, M. Esters & S. Curtarolo.
3. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at the North Carolina State University, Raleigh, North Carolina (Mar 12, 2019). **Co-Organizers**: C. Toher, D. Hicks, E. Gossett & M. J. Brenner.

2. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Co-Organizer and Presenter** at Carnegie Mellon University, Pittsburgh, Pennsylvania (Jan 21, 2019). **Co-Organizers**: D. Hicks & E. Gossett.

2018

1. *AFLOW School: Integrated infrastructure for computational materials discovery*. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Machine Learning Quantum Materials of the University of Maryland/NIST/Moore Foundation, Institute for Bioscience & Biotechnology Research in Gaithersburg, Maryland (Aug 02, 2018). **Co-Presenters**: C. Toher & D. Hicks.
   - website: https://www.nanocenter.umd.edu/events/mlmr/program/

Professional Service

Journal Reviewer	2018–2026	68 manuscripts for 34 journals; 57 since 2022, listed by year
	2026	12 manuscripts: Advanced Materials, Communications Materials, International Journal of Applied Glass Science, Journal of Chemical Theory and Computation, Journal of the American Chemical Society, Machine Learning for Computational Science and Engineering, Nature Communications, npj Computational Materials, Physical Review B, The Journal of Physical Chemistry
	2025	10 manuscripts: Journal of the American Chemical Society, Nature Communications, npj Computational Materials, Physical Review B, Science Advances, Scientific Reports
	2024	9 manuscripts: Acta Crystallographica Section B, Advanced Materials, Applied Physics Letters, Chemistry of Materials, Nature Computational Science, Nature Synthesis, Proceedings of the National Academy of Sciences, Scientific Reports, Scripta Materialia
	2023	15 manuscripts: ACS Nano, Acta Materialia, Computer Physics Communications, Nature Communications, npj Computational Materials, Scientific Reports, Scripta Materialia
	2022	11 manuscripts: ACS Catalysis, ACS Omega, Acta Materialia, APL Materials, Chemistry of Materials, Frontiers in Energy Research, Materials Letters, npj Computational Materials, Scientific Data
Panel and Ad Hoc Reviewer	2023, 2025	National Science Foundation (DMREF, DMR, CDSE & RUI)
Reviewer	2021, 2023, 2025	Department of Energy, Basic Energy Sciences
Reviewer	2021, 2026	American Chemical Society Petroleum Research Fund
Reviewer	2022	Stanford Synchrotron Radiation Lightsource (beamtime proposals)
Reviewer	2024, 2025	Johns Hopkins University, Discovery Award
Reviewer	2024, 2026	Johns Hopkins University, IDIES Seed Funding and Data Science and AI Institute Trusted Datasets
Guest Editor and Roadmap Organizer	2026–present	JPhys Energy Roadmap, "AI-Enabled Resilient Energy Systems: Technologies, Intelligence, and Security"
Guest Editor	2024–2025	*Materials Letters* Virtual Special Issue, "High-Entropy Materials"
Member	2022–present	The Minerals, Metals & Materials Society
Member	2014–present	American Physical Society

Participant	2018–2024	Open Databases Integration for Materials Design (OPTiMaDe) Workshop

Press and News Releases

JHU Applied Physics Laboratory	Aug 13, 2026	*SURPASS Projects Tackle the 'Beyond Possible'* • This press release is featured on Office of the Vice Provost for Research at JHU. jhuapl.edu/news/news-releases/260813-surpass
JHU Engineering News	Jun 30, 2026	*Laboratory to Launchpad Helps Researchers Navigate the Path from Discovery to Impact* • This press release is featured on JHU Energy Institute News. engineering.jhu.edu/materials/news/laboratory-to-launchpad-helps-researchers-navigate-the-path-from-discovery-to-impact
JHU Engineering News	Jun 22, 2026	*MatSci faculty win OVPR awards* • This press release is featured on Office of the Vice Provost for Research at JHU. engineering.jhu.edu/materials/news/matsci-faculty-win-ovpr-awards
JHU Energy Institute News	Feb 05, 2026	*Corey Oses Awarded ARPA-E IGNIITE Funding for AI-Driven Fusion Materials Research* energyinstitute.jhu.edu/corey-oses-awarded-arpa-e-igniite-funding-for-ai-driven-fusion-materials-research
The New Indian Express	Jan 24, 2026	*Wings of fire to fly high in AI* newindianexpress.com/good-news/2026/Jan/25/wings-of-fire-to-fly-high-in-ai
JHU Energy Institute News	Dec 17, 2025	*The Platinum Problem* • This press release is featured on JHU Engineering News. energyinstitute.jhu.edu/the-platinum-problem
JHU Engineering News	Jun 16, 2025	*Oses and Team Win 2025 Nexus Award* engineering.jhu.edu/materials/news/oses-and-team-win-2025-nexus-award
JHU Energy Institute News	Oct 16, 2024	*Corey Oses Earns ISMM Early-Career Investigator Award* • This press release is featured on JHU Engineering News. energyinstitute.jhu.edu/corey-oses-earns-ismm-early-career-investigator-award
JHU Energy Institute News	Jun 12, 2024	*Team Co-Led by Corey Oses Receives SURPASS Award* • This press release is featured on JHU Engineering News. energyinstitute.jhu.edu/team-co-led-by-corey-oses-receives-surpass-award
JHU Engineering News	Jun 21, 2023	*Unlocking the potential of thermoelectric materials* engineering.jhu.edu/news/unlocking-the-potential-of-thermoelectric-materials
Duke Engineering News	Oct 11, 2022	*Heat-Proof Chaotic Carbides Could Revolutionize Aerospace Technology* pratt.duke.edu/about/news/heat-proof-chaotic-carbides-could-revolutionize-aerospace-technology
White House Office of Science & Technology Policy	Nov 18, 2021	*Featured Vignette in the November 2021 Materials Genome Initiative Strategic Plan (page 9)* mgi.gov/sites/default/files/documents/MGI-2021-Strategic-Plan.pdf
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